



THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA

Student Number Family Name First Name

Do not distribute this document

School of Mathematics & Physics
EXAMINATION

Semester One Final Examinations, 2020

MATH1061 Discrete Mathematics

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes (+ additional 30 mins encompassing reading time, and time to scan and upload solutions). You must commence your exam at the time listed in your personalised timetable. The exam will remain open **only** for the duration shown.

Materials Permitted While Completing the Exam: You may use any online or physical material you wish. However, you have to write solutions in your own words (i.e. copying is considered cheating). You are NOT allowed to contact another human being directly or indirectly regarding this test. In particular, posting questions to chat forums is considered cheating.

Instructions to Students: Answer all questions. Upload as a **SINGLE PDF FILE**. You can print the exam and write on the exam paper, or write your answers on blank paper, or write electronically on a suitable device. Scan or photograph your work if necessary and upload your answers to Blackboard as a single pdf file.

Who to Contact Since students may not all undertake the online exam at the same time, or in the same time zone, and that some questions may be randomised, responding to student queries and/or relaying corrections to exam content during the exam will not be feasible. Course coordinators will not be able to respond to academic queries during the exam.

If you have any concerns or queries about a particular question, or need to make any assumptions to answer the question, state these at the start of your solution to that question. You may also include queries you may have made with respect to a particular question, should you have been able to 'raise your hand' in an examination room.

If you experience any technical difficulties during the exam, contact the [Library AskUs](https://web.library.uq.edu.au/contact-us) service via the Live Chat or Phone for advice (<https://web.library.uq.edu.au/contact-us> - open 7:00am – 10:00pm AEST every day during the final exam period). You should ask the library staff for an email documenting the advice provided so you can provide this to the course coordinator.

Certification (must be signed before submission):

I certify that my submitted answers are entirely my own work and that I have neither given nor received any unauthorised assistance on this assessment item.

Signed: _____ Date: _____

1. Let the cyclic group $\{[0], [1], [2], \dots, [n-1]\}$ be denoted by $\mathbb{Z}/n\mathbb{Z}$. Consider the following statement: for every positive integer n and every x in $\mathbb{Z}/n\mathbb{Z}$, there exists $y \in \mathbb{Z}/n\mathbb{Z}$ such that $xy = [1]$.
- (a) (2 marks) Write the negation of this statement.

Solution: The negation is: There exists a positive integer n and $x \in \mathbb{Z}/n\mathbb{Z}$ such that for all $y \in \mathbb{Z}/n\mathbb{Z}$, $xy \neq 1$.

No part mark for this question.

- (b) (2 marks) Is the original statement true or false? Justify your answer.

Solution: The negation is the correct statement. For instance, take $n = 4$ and $x = [2]$. Then x has no multiplicative inverse in $\mathbb{Z}/4\mathbb{Z}$.

No part mark. If they get it right, they get 2. If not they get 0.

Aside for graders

There may be many correct solutions possible for each of the following questions. As a general rule, if an answer is correct, it gets full mark, regardless of the method. The usual notation is n for an integer and p for a prime. I have changed most integers n 's to x 's and p 's to q 's so that it is not easy to Google. But in the solutions, I have reverted to the common notations. Translation should be trivial.

2. (4 marks) Let x be a fixed positive integer. Is it possible to have a graph G with $4x + 1$ vertices such that G has a vertex of degree d for all $d = 1, 2, \dots, 4x + 1$? Justify your answer.
(Note: The graph G does not need to be simple.)

Solution: No. If such a graph existed, then the vertices of degree 1, 3, ..., $4x + 1$ are all odd. There are $2x + 1$ such vertices. Thus, such a graph would have odd number of odd vertices, which is a contradiction to the Handshake Theorem.

2 marks for answering No.

2 marks for providing correct reasoning.

3. (4 marks) Let q be an odd prime. Let x be an integer satisfying $1 < x < q$. Prove that

$$q \mid \binom{q}{x}.$$

Solution: Recall that that

$$\binom{p}{n} = \frac{p!}{n!(p-n)!}.$$

Clearly p divides the numerator. Thus, it is sufficient to show that p does not divide the denominator. By assumption both n and $p - n$ are strictly between 1 and p . Thus, p does not divide any number less than or equal to n or $p - n$. As p is a prime, it follows that p does not divide $n!$ or $(p - n)!$ or their product.

As there are many possible solutions, no grading scheme is provided. Use your own judgement.

4. (4 marks) Consider the function $f : \mathbb{R} - \{\pm 1\} \rightarrow \mathbb{R}$ defined by

$$f(x) = \frac{1}{x^2 - 1}.$$

Determine the range of f . Justify your answer.

Solution: First observe that 0 is not in the range. Next, if $y \in \mathbb{R} - \{0\}$, then

$$y = f(x) = \frac{1}{x^2 - 1} \implies x^2 - 1 = \frac{1}{y} \implies x^2 = 1 + \frac{1}{y}.$$

This has a solution if and only if

$$1 + \frac{1}{y} \geq 0 \iff \frac{1}{y} \geq -1 \iff y > 0 \text{ or } y \leq -1.$$

Thus, the range is $(-\infty, -1] \cup (0, \infty)$.

2 marks for getting the answer correct.

2 marks for correct justification.

5. (4 marks) Let x be a fixed but arbitrary integer. Do there exist infinitely many integers y such that

$$y^3 \equiv x^{12} \pmod{2019}?$$

Justify your answer.

Solution: We claim that there exists infinitely many m such that 2019 divides $n^{12} - m^3$. First observe that $n^4 - m$ divides $n^{12} - m^3$. To see this note that $x - y$ divides $x^3 - y^3$ as $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$. Now set $x = n^4$ and $y = m$. Thus it is sufficient prove that there exists infinitely many m such that $2019 \mid (n^4 - m)$. But this is obvious, since we can take m to be any element of the set

$$n^4, n^4 + 2019, n^4 + 2 \times 2019, \dots$$

Reformulation of the solution: Set $m = n^4 + 2019k$ where k is any integer. Now observe that

$$m^3 = (n^4 + 2019k)^3 = n^{12} + 3n^8(2019k) + 3n^4(2019k)^2 + (2019k)^3 \equiv n^{12} \pmod{2019}.$$

As there are many possible solutions, no grading scheme is provided. Use your own judgement.

6. Let S be a finite set and let $\mathcal{P}(S)$ denote the set of all subsets of S . Define a relation on $\mathcal{P}(S)$ by declaring that two subsets A and B are related if A and B have the same number of elements.
- (a) (2 marks) Prove that this is an equivalence relation.

(b) (2 marks) Determine the equivalence classes.

(c) (2 marks) Determine the number of elements in each equivalence class.

Solution: Clearly $|A| = |A|$ and if $|A| = |B|$ then $|B| = |A|$. Moreover, if $|A| = |B|$ and $|B| = |C|$, then $|A| = |C|$. Thus, we have an equivalence relation. The equivalence classes are

$$[1], [2], \dots, [n]$$

where $[k]$ denotes the set of subsets of S which have exactly k elements. The equivalence class $[k]$ has $\binom{n}{k}$ elements.

For each part, they get 2 if they answer correctly and 0 otherwise.

7. (6 marks) Consider the sequence $\{a_n\}_{n \geq 1}$ defined by $a_1 = 3$, $a_2 = 8$, and

$$a_n = 4(a_{n-1} - a_{n-2}), \quad \forall n \geq 3.$$

Prove that $a_n = 2^{n-1}(n+2)$ for all $n \geq 1$.

Solution: We prove this by strong induction. Note that

$$a_1 = 3 = 2^0 \cdot 3 = 2^{1-1}(1+2),$$

and

$$a_2 = 8 = 2 \cdot 4 = 2^{2-1}(2+2).$$

Thus, our base cases are covered. Suppose the hypothesis is true for all $k < n$. Then,

$$a_n = 4a_{n-1} - 4a_{n-2} = 4(2^{n-2}(n+1) - 2^{n-3}n) = 4 \cdot 2^{n-3}(2n+2-n) = 2^{n-1}(n+2).$$

Induction base is worth 2 marks. Induction step is worth 4.

8. (8 marks) A group of 3 friends go to a concert with total attendance 30. At the end of the concert, the coat check system crashes, so they return people's coats at random. Determine the probability that at least one of the 3 friends obtains their own coat. Write your answer in the fraction form $\frac{a}{b}$ where $\gcd(a, b) = 1$.

Solution: Let $\{a_1, a_2, a_3\}$ denote the three friends. The total number of permutations is $30!$. If we assume that a_1 gets their own coat back, then the total number of permutations is $29!$. Similarly, if we assume that a_1 and a_2 get their coats, then the number of permutations is $28!$ and if we assume all three get their coats, then it is $27!$. Now the Inclusion/Exclusion principle gives:

$$\begin{aligned}
 P &= P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_2 \cap A_3) - P(A_3 \cap A_1) + P(A_1 \cap A_2 \cap A_3) \\
 &= \frac{1}{30!}(3 \cdot 29! - 3 \cdot 28! + 27!) = \frac{27!(3 \cdot 29 \cdot 28 - 3 \cdot 28 + 1)}{30!} = \frac{3 \cdot 29 \cdot 28 - 3 \cdot 28 + 1}{30 \cdot 29 \cdot 28} \\
 &= \frac{2436 - 84 + 1}{24360} = \frac{2353}{24360}.
 \end{aligned}$$

4 marks for the correct fraction.

4 marks for the correct explanation.

9. (10 marks) Define a binary operation on $\mathbb{R}^2 - \{(0, 0)\}$ by

$$(a, b) \cdot (c, d) = (ac - bd, ad + bc).$$

Prove that $(\mathbb{R}^2 - \{0\}, \cdot)$ is an abelian group.

(You do not need to prove that the operation is closed.)

Solution: Associativity (4 marks): note that

$$(a, b) \cdot ((c, d) \cdot (x, y)) = (a, b) \cdot (cx - dy, cy + dx) = (acx - ady - bcy - bdx, acy + adx + bcx - bdy)$$

equals

$$((a, b) \cdot (c, d)) \cdot (x, y) = (ac - bd, ad + bc) \cdot (x, y) = (acx - bdx - ady - bcy, acy - bdy + adx + bcx).$$

Identity (2 marks) We claim $(1, 0)$ is the identity. This follows from

$$(a, b) \cdot (1, 0) = (a, b) = (1, 0) \cdot (a, b).$$

Inverse (2 marks): We have

$$(a, b) \cdot (a, -b) = (a, -b) \cdot (a, b) = (a^2 + b^2, 0).$$

Thus, every element (a, b) has an inverse; namely, $\frac{1}{a^2+b^2}(a, -b)$.

Abelian (2 marks) Finally, the group is abelian because

$$(a, b) \cdot (c, d) = (ac - bd, ad + bc) = (ca - db, cb + da) = (c, d) \cdot (a, b).$$

Sketch of an alternative solution: Identify $\mathbb{R}^2$ with $\mathbb{C}$ via the map $(a, b) \mapsto a + bi$. Then T gets identified with $\mathbb{C}^\times$ and the above operation is the natural multiplication on $\mathbb{C}^\times$.

10. (10 marks) Let x be an integer bigger than 4. Prove that x is prime if and only if x does not divide $(x-1)!$.

Solution: [$\implies$ worth 4 marks] First observe that n does not divide any k satisfying $1 \leq k < n$. Now suppose n is a prime. Then this means that n does not divide the product $1.2... (n-1) = (n-1)!$. This proves one direction. For converse (worth 6 marks), we offer three solution:

Ben's quick solution: If n is composite then write $n = ab$ with $a, b < n$. If $a \neq b$ then both a and b appear as different factors of $(n-1)!$. If $a = b$ then (since $n > 4$) both a and $2a = 2b$ appear as different factors of $(n-1)!$. Either way you get $n = ab \mid (n-1)!$.

Another solution: Write $n = ab$ with $a, b \geq 2$. Note that as $n > 4$, at least one of a or b is strictly bigger than 2. Thus,

$$a + b = n/b + n/a < n/2 + n/2 = n \implies a + b \leq n - 1 \implies (a + b)! \mid (n - 1)!.$$

As $\binom{a+b}{a}$ is an integer, we obtain

$$n = ab \mid a!b! \mid (a + b)! \mid (n - 1)!.$$

My terrible solution: By the fundamental theorem of arithmetic, $n = p_1^{a_1} \dots p_k^{a_k}$. Suppose $n > 4$ is composite. To show $n \mid (n-1)!$, it suffices to prove $p_i^{a_i} \mid (n-1)!$ for every i . Henceforth, we fix i and write a for a_i and p for p_i . The statement is clear if $a = 1$ because $p < n$ and therefore, $(n-1)!$ is divisible by p . For $p = 2$, the statement can be proved as follows. Note that we have $\lfloor (n-1)/2 \rfloor$ even integers less than or equal to $n-1$. Thus, it suffices to show that

$$2^{\lfloor (n-1)/2 \rfloor} \geq n-1, \quad \forall n > 4.$$

This can be proved by induction. Finally we prove the statement for all odd primes. First, we prove by induction that $p^a > ap$ for all $p > 4$. It follows that all numbers

$$p, 2p, \dots, ap$$

are all less than $(n-1)$. Thus, $(n-1)!$ is divisible by p^a as required.